MA4N1 Sylow Theorems Project Outline

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Introduction

In this project, we intend to formalise the Sylow Theorems, which we will take to be the following four statements.

Theorem (Sylow Theorems)**.** Let G be a finite group, and let p be prime. Then the following conditions hold:

1. G has at least one Sylow p-subgroup.
2. All of the Sylow p-subgroups of G are conjugate.
3. Any p-subgroup of G is contained in a Sylow p-subgroup of G.
4. The number of Sylow p-subgroups is congruent to 1 modulo p.

In this project, we will only be importing Mathlib libraries that cover the necessary definitions (i.e. Group, Subgroup, Group action, etc.) and the basic properties of these definitions. Any further properties of groups, such as the Orbit-Stabiliser Theorem and Lagrange's Theorem, we will prove ourselves.

Outline

The main focus of this project will be formalising Wielandt's proof[1] of the fact that $|\mathrm{Syl}_p(G)| \equiv 1 \pmod p$, which is equivalent to the first and fourth Sylow Theorems holding. In order to formalise this proof, we need primarily the two key theorems named in the previous section, a number theoretic lemma outlined below and several innocuous results that require precise formalisation.

The following section outlines how we have split the project into manageable parts and how each member of the group will contribute to the final project. We plan to formalise the statements of each lemma to create a "skeleton" proof of our goal; each lemma will have a placeholder "sorry" as its proof, so that the lemmas can be used by other team members. This will allow us to concurrently work on proving our assigned lemmas. We will then fill in the placeholder "sorry" for our respective lemmas with our completed proofs.

This division is a rough outline, and is subject to change depending on the difficulty and size of each section.

Jia

Formalise the Orbit–Stabiliser Theorem and its corollary describing the partition of a set into orbits.

Theorem. Let the finite group G act on X, and let $x \in X$, then $|G| = |Orb_G(x)||Stab_G(x)|$.

Corollary. Let G be a finite group acting on a set X, and let $x \in X$. Then

(i) Either $Orb_G(x) = Orb_G(y)$ or $Orb_G(x) \bigcap Orb_G(y) = \emptyset$.

[1] See pages 32-35 of *MA3K4 Introduction to Group Theory Complete Lecture Notes* on the Introduction to Group Theory moodle page

(ii) The set $\{Orb_G(x) : x \in X\}$ forms a partition of X.

(iii) $|Orb_G(x)|$ divides $|G|$.

Reece

Formalise Lagrange's Theorem by proving lemmas regarding cosets such as:

- For $g_1, g_2 \in G$, then $g_1 \sim_H g_2$ if $g_1H = g_2H$ is an equivalence relation
- Let $\{g_iH : i \in I\}$ be the set of $\sim_H$-equivalence classes in G (I here is an index set). Then $G = \bigsqcup_{i \in I} g_iH$ is the disjoint union of the sets g_iH, for $i \in I$.

Ben

Formalise the following number-theoretic lemmas.

Lemma. Let p be prime, and let n and m be integers with $\gcd(m,p) = 1$. Then

(i) p divides $\binom{p}{i}$ for all $1 \leq i \leq p-1$; and

(ii) $\binom{p^n m}{p^n} \equiv m \pmod{p}$

I will attempt to formalise this following the proof from MA3K4 Introduction to Group Theory.

The proof of (i) involves some prime divisibility results. The proof of (ii) involves the binomial coefficients of the polynomials $(1+X)^{pm}$ and $(1+X^p)^m$ in the field $\mathbb{Z}/p\mathbb{Z}$.

Beth

Formalise statements concerning bijections and their properties, as required in the proofs of Lagrange's Theorem and the Orbit-Stabiliser Theorem.

- If there exists a bijection between finite groups G and H, then $|G| = |H|$
- If $V_1, V_2, ..., V_n$ are disjoint finite sets, then the disjoint union $V = V_1 \sqcup V_2 \sqcup ... \sqcup V_n$ has $|V| = \Sigma_{i=1}^{n} |V_i|$

Possible Extension

If time allows, we will prove the second and third statements of the Sylow Theorems, which follow from a separate lemma regarding normal subgroups.

Within the outline there are two parts which do not fall under the scope of group theory, namely the number theoretic lemma and the equivalence relation knowledge required to prove Lagrange's Theorem. The statements of these facts are essential to our proof, but as the project is principally a group theory project they may not be formalised to the same extent. However, if there is time to do so, we intend to formalise these sections fully.